

THESIS PROPOSAL

THESIS TITLE: BUILDING A STATISTICALLY VALIDATED DECISION SUPPORT SYSTEM FOR FOREIGN EXCHANGE TRADING WITH CALIBRATED UNCERTAINTY AND RISK CONTROL

Advisor: ASSOCIATE PROFESSOR NGUYEN DINH THUAN

Implementation period: 09/2026 to 12/2026 (16 weeks)

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Introduction:

The foreign exchange (FX) market is the largest and most liquid financial market in the world, and a large number of decision support tools are marketed to retail participants. Such tools typically display a directional forecast, a set of chart-pattern signals, and a qualitative confidence label. The empirical foundation for these outputs is, however, considerably weaker than their presentation suggests. Daily exchange rate returns have resisted prediction since the benchmark result of Meese and Rogoff [1], and large-scale replications of technical trading rules in currencies report that in-sample profitability largely disappears out of sample and that residual abnormal returns are absorbed by a single time-series momentum factor [2], [3].

The central methodological difficulty is not model capacity but **inference under a low signal-to-noise ratio combined with a large hypothesis space**. A system that evaluates thousands of candidate rules, feature sets, or model configurations will identify many nominally significant relationships purely by chance. Unless the hypothesis space is enumerated in advance and the family-wise error rate is controlled — for example by resampling-based step-down procedures [4] or by tests of superior predictive ability over an entire model family [5] — the resulting system reports discoveries that do not survive replication [6]. A second and less frequently addressed difficulty concerns **negative findings**: reporting that no effect was detected is uninformative unless the minimum detectable effect size (MDES) of the testing procedure is also reported, because "there is nothing there" and "the procedure could not have found it" are otherwise indistinguishable.

A third observation motivates the design adopted here. Predictability in FX is not a single property: it should be decomposed by **target axis**. The *direction* axis (will the price rise or fall) and the *magnitude* axis (how large will the movement be) have very different empirical status. The

heterogeneous autoregressive (HAR) model of realized volatility [7] and its extensions incorporating measurement-error correction [8] and jump/semivariance components [9] provide well-established magnitude forecasts, whereas directional skill remains contested. Recent comparative studies further report that nonlinear machine-learning models do not statistically outperform linear specifications for realized volatility once multiple testing is accounted for [10], and that time-series foundation models do not consistently beat a log-HAR benchmark [11]. A decision support system should therefore state explicitly *on which axis* it claims skill, and should demonstrate that claim under a fixed evaluation protocol rather than asserting it.

Preliminary work carried out by the group has established the feasibility and the empirical baseline for this design, and has already progressed through three staged, pre-registered information layers, each closed with a written decision record before the next was opened. A complete data pipeline has been constructed for six major currency pairs (EUR/USD, GBP/USD, USD/JPY, AUD/USD, USD/CAD, USD/CHF) over 2010–2025, aggregating approximately 34.9 million one-minute bars into daily bars and five-minute realized measures; a production volatility layer, a three-class probability layer, and a risk layer have been implemented end to end and are served through an operating web application updated on a daily schedule. Under a fixed 70/15/15 temporal split, the extended HAR specification attains a test-set QLIKE of 0.1585 against 0.2172 for the previous baseline, while fourteen machine-learning and deep-learning alternatives, two zero-shot time-series foundation models, and 509 forecast combinations fail to displace it from the Model Confidence Set [12]. On the direction axis, 8,652 fully enumerated rule hypotheses across twelve independent discovery branches — spanning historical price patterns and, in a subsequent closed phase, four independent representations of official central-bank communication — yield zero survivors after family-wise error control, with a measured detection power corresponding to a lift of 1.20. A further pre-registered investigation of exogenous macroeconomic and cross-asset variables under a causal-versus-validation feature-selection ablation has also been completed and frozen, with its sealed-set evaluation deliberately withheld because the pre-specified decision rule for opening it was not met.

This thesis therefore addresses a research problem that is methodological as much as it is predictive: **how to construct a decision support system for FX whose every displayed quantity is traceable to a measurement under a pre-registered evaluation protocol, which states the axis on which it has skill, which quantifies the strength of its negative findings, and which converts calibrated uncertainty into actionable risk limits.** The closed phases established what does and does not carry information; the remaining work consists of completing a small number of explicitly identified residual validation items within the existing protocol (rather than reopening its settled conclusions), performing the single sealed-set evaluation, and finalising the system and its documentation.

Research Objectives:

The primary objective of this research is to design, implement, and empirically validate a decision support system for foreign exchange trading in which forecast skill is established per target axis under strict multiple-testing control, uncertainty is communicated through calibrated conformal prediction sets, and position sizing is constrained by an estimated probability of ruin under explicitly

stated assumptions.

Consistent with this objective, three groups of quantities are kept conceptually distinct throughout the study: **predictive axes** (direction, magnitude, and tail-event probability, each scored against its own baseline and its own proper scoring rule), **calibration diagnostics** (coverage and reliability of the probabilistic and conformal outputs), and the **decision layer** (Value-at-Risk, Expected Shortfall, stop-out probability, and position sizing, which consume the predictive and calibration outputs but are not themselves treated as an additional predictive claim).

Specific objectives include:

- **Establishing axis-specific predictability.** Determine, under a single fixed evaluation protocol, on which of the three target axes — direction, magnitude, and risk — out-of-sample predictive skill can be demonstrated for daily FX, and report the result on each axis separately rather than through a single aggregate score.
- **Benchmarking model complexity against a well-specified econometric baseline.** Compare linear HAR-type specifications with tree-based ensembles, recurrent and attention-based neural architectures, tabular and time-series foundation models, and systematic forecast combinations, using proper scoring rules and formal model-comparison tests.
- **Quantifying the incremental value of additional information layers.** Evaluate whether textual central-bank communication, macroeconomic and cross-asset exogenous variables, and daily candlestick geometry add measurable predictive value beyond price history, and determine whether *causal/temporal filtering* of such variables outperforms ordinary validation-based feature selection.
- **Delivering calibrated uncertainty.** Implement and validate adaptive conformal prediction so that the prediction sets displayed to the user are designed to maintain target coverage under distribution shift, verify this with empirical and regime-conditional coverage under the assumptions of the selected conformal procedure, and assess calibration with proper scoring rules and reliability diagnostics.
- **Converting forecasts into risk-controlled decisions.** Produce a per-session decision record comprising forecast volatility, recommended stop distance, probability of stop-out by holding horizon, Value-at-Risk and Expected Shortfall with formal backtests, and a ruin-constrained position size adjusted for portfolio correlation and measured slippage.
- **Reporting negative findings with quantified power.** For every negative result, report the minimum detectable effect size of the corresponding testing procedure, so that each negative claim is bounded rather than open-ended.
- **Implementing the complete system.** Deliver a reproducible web-based decision support system in which every displayed number is traceable to a specific measurement and its supporting evidence.

Research Scope:

This research focuses on **daily-horizon decision support for six major currency pairs involving the U.S. dollar**: EUR/USD, GBP/USD, USD/JPY, AUD/USD, USD/CAD, and USD/CHF (the dollar is the quote currency in the first three pairs and the base currency in the last three). The

development sample covers 2010–2025 and is partitioned temporally into a training segment, a validation segment used only for model and hyper-parameter selection, and a held-out development-test segment that is scored once and is distinct from the sealed evaluation set described below.

A **sealed evaluation set** — six cross pairs not used anywhere in development, namely EUR/GBP, EUR/JPY, GBP/JPY, AUD/JPY, EUR/CHF, and NZD/USD, together with 2026 data as available at the pre-specified cutoff date for all twelve pairs — is reserved and is opened exactly once, after the final configuration has been frozen in writing; because the sealed set includes calendar year 2026 while the study concludes before that year ends, "2026" refers to whatever portion of the year has been collected through the continuously operating daily pipeline by the cutoff date, not the completed calendar year. No retuning is permitted after the sealed set is opened.

The primary data source is aggregated one-minute historical bar data, from which daily bars and five-minute realized measures (realized variance, realized quarticity, bipower variation, and signed semivariances) are computed. Supplementary sources include an official central-bank meeting and macroeconomic release calendar, daily exogenous series obtained from a public economic data service, published monetary-policy surprise measures, interest-rate differentials, and transaction-cost measurements derived from tick data with bid–ask quotations.

Forecast horizons considered are one, five, and twenty trading sessions. The one-session horizon is the operational default; longer horizons are reported with their measured skill level rather than presented as equivalent.

The study is explicitly **not** an attempt to construct a profitable automated trading strategy. Order execution, order-book microstructure modelling, market impact of large orders, intraday trade timing below the daily decision, and portfolio optimisation across asset classes other than FX are outside the scope. Statements about causality are limited to temporal predictive relationships in the Granger sense and are not presented as structural causal claims.

Research Subjects:

The primary research subject is **the statistical validation of decision support outputs for foreign exchange trading under a low signal-to-noise ratio**.

The study examines how forecast skill can be attributed to a specific target axis and to a specific information layer; how multiple-testing control, power analysis, leakage protection, and sealed-set validation interact to produce claims that survive replication; how calibrated probabilistic forecasts and conformal prediction sets can be converted into risk limits with a measured probability of ruin; and how such evidence should be presented in a decision support interface so that the user can distinguish what the system measures from what it merely assumes.

Research Methodology:

The research is conducted through the following main activities.

Literature Review

- Review realized-volatility modelling, covering the HAR family and its measurement-error, jump, and semivariance extensions, and recent comparative evidence on machine learning and

foundation models for volatility forecasting.

- Review the evidence on FX directional predictability, including large-scale technical-rule studies with data-snooping control, and the momentum-factor explanation of residual rule profitability.
- Review statistically sound pattern discovery, family-wise error control, superior predictive ability testing, and model confidence sets.
- Review conformal prediction under distribution shift, proper scoring rules, and calibration diagnostics.
- Review Value-at-Risk and Expected Shortfall backtesting, and growth-optimal position sizing under a ruin constraint.

Data Preparation and Protocol Fixing

- Construct daily bars and five-minute realized measures from one-minute source data; verify contiguity, session conventions, and time-zone handling.
- Fix the temporal partition, the evaluation metrics per axis, the multiple-testing procedure, and the sealed-set policy in a written record **before** any scoring, and record every configuration evaluated on the development sample.
- Implement a permanent anti-leakage protocol: every derived feature must satisfy the self-test that truncating all information revealed after time t leaves its value at t unchanged.

Baseline Establishment and Model Comparison

- Establish the extended HAR specification as the production magnitude baseline and an online expert-combination layer [13] as the three-class probability baseline.
- Compare against linear regularised models, gradient-boosted trees, recurrent and attention-based networks, a tabular foundation model [14], zero-shot time-series foundation models [11], and systematic forecast combinations, all under an identical information set, refit frequency, and variance-conversion procedure.
- Evaluate with scale-invariant QLIKE [15], the continuous ranked probability score, and the logarithmic and Brier scores [16], and assess significance with Diebold–Mariano tests [17] and Model Confidence Sets [12].

Pattern, Exogenous, and Causal Analysis

- Enumerate the hypothesis space in advance for every discovery family — symbolic sequence patterns, motif and matrix-profile analysis, interpretable rule learning, regime models, textual central-bank features, exogenous macro and cross-asset variables, and daily candlestick geometry — and apply step-down max- T resampling with block-preserving null models [4]. Remaining work in this family is limited to completing pre-registered evaluations at horizons and axes not yet scored (Section on the Research Plan); no new discovery family will be introduced after the evaluation protocol has been frozen.
- Condition every candidate on the strongest available null: the production volatility forecast for the magnitude axis, and time-series momentum for the direction axis [3].
- For the exogenous layer, contrast three feature-selection regimes under an identical model class and parameter budget: all declared variables, validation-selected variables, and variables filtered by temporal-causal testing [18] with a cross-pair and cross-year stability screen, in order to

isolate the contribution of causal filtering itself.

- Screen surviving candidates for sign consistency across pairs and across years, and require leave-one-pair-out transfer before any candidate is admitted.

Calibration, Risk, and Decision Layer

- Implement adaptive conformal inference with Mondrian stratification by volatility regime [19], [20], and verify empirical coverage overall and conditional on regime.
- Estimate Value-at-Risk and Expected Shortfall from empirical quantiles of standardised returns scaled by the volatility forecast, and backtest per pair using unconditional coverage [21], independence and conditional coverage [22], and dynamic quantile tests [23]; assess the joint (VaR, ES) pair with a strictly consistent scoring function [24].
- Compute position size as the minimum of a growth-optimal fraction [25] and a cap constrained by an estimated probability of ruin, adjusted by volatility regime, drawdown state, a portfolio correlation factor, and a slippage coefficient measured from observed stop-loss executions; the ruin estimate is reported together with the return, dependence, and sizing assumptions on which it is conditioned.

Power Analysis and Sealed Validation

- For every negative finding, inject synthetic effects of known magnitude through the actual testing funnel and report the effect size detected with 80% power, together with a negative control establishing the false-positive rate under pure noise.
- Freeze the final configuration in writing, then open the sealed evaluation set exactly once and report the resulting numbers without modification.

System Implementation and Analysis

- Maintain, throughout the study rather than only at its conclusion, an operating end-to-end product path — data ingestion, production forecast artefact, application programming interface, and a minimal user interface — so that each research layer is integrated as a versioned artefact update instead of being connected for the first time late in the project; dedicate the final weeks to hardening this system with a computation layer, a versioned forecast-artefact interface, and a user interface that presents per-axis skill, conformal prediction sets, the event calendar with measured historical responses, and the risk decision record with full provenance.
- Analyse failure cases and limitations, including pairs that fail tail backtests and horizons at which calibration degrades, and report them within the interface rather than omitting them.

Expected Results and Contributions:

The expected outcomes of this research include:

- **A validated decision support system for FX** that issues, for each session and each pair, a volatility forecast, a calibrated three-class probability distribution with a conformal prediction set, an event-risk assessment, and a ruin-constrained position-sizing recommendation, with every displayed quantity traceable to a specific measurement.
- **An axis-resolved characterisation of daily FX predictability**, establishing where skill exists and where it does not, under a single protocol applied uniformly across all information layers.

Results obtained and frozen to date indicate positive and statistically significant skill on the magnitude axis and no detectable skill on the direction axis across twelve independent discovery branches, including four independent representations of price history and of official central-bank communication; the thesis will complete this analysis at the remaining horizons and confirm it on the sealed evaluation set, which has not yet been opened.

- **A systematic benchmark of model complexity for FX volatility forecasting**, covering linear, tree-based, recurrent, attention-based, tabular-foundation and time-series-foundation models, together with exhaustive forecast combinations, evaluated with proper scoring rules and model confidence sets.
- **An empirical comparison of causal filtering against ordinary feature selection.** A completed, frozen ablation (not yet confirmed on sealed data) shows that variables filtered by temporal-causal testing with a stability screen degrade substantially less out of sample than variables selected by validation performance, with the mechanism identified and measured as distribution drift in the selected macroeconomic variables. This comparison appears to be underexplored in the FX forecasting literature.
- **Power-bounded negative evidence.** Each negative finding will be reported together with the minimum effect size detectable by the corresponding procedure, converting statements of the form "no relationship was found" into bounded claims of the form "no relationship stronger than X exists in these data".
- **Three applied contributions to risk-aware decision support:** a portfolio correlation factor for ruin-constrained sizing, quantifying the gap between the nominal and the realised probability of ruin when multiple correlated positions are opened simultaneously; a holding-horizon table on the decision record, addressing the systematic misreading of single-session stop-out probabilities; and a slippage coefficient estimated from observed stop-loss executions rather than assumed.
- **A reproducible research record** containing the full data dictionary, the sealed-set protocol, the registry of every model configuration and hypothesis evaluated, and reproduction commands for every reported number, including all negative results.

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Research Plan and Timeline:

Research timeline: 16 weeks.

No.	Assignments	Timeline
1	– Consolidate the data pipeline and verify contiguity, session conventions and time-zone handling. – Fix the evaluation protocol: temporal partition, per-axis metrics, multiple-testing procedure, and sealed-set policy. – Reproduce all baseline results from a clean checkout.	Week 1–2
2	– Complete the pattern-discovery families remaining at the five- and twenty-session horizons. – Apply superior predictive ability testing at the family level across all discovery branches. – Document the enumerated hypothesis space in full.	Week 3–4
3	– Conduct minimum detectable effect size analysis for every negative finding. – Validate the discovery funnel with negative controls and known-effect injection. – Formalise the reporting template for power-bounded negative claims.	Week 5–6

4	– Address tail-risk failures for the two pairs that do not pass Value-at-Risk and Expected Shortfall backtests at the 99% level, using conditional and extreme-value approaches. – Recalibrate the twenty-session horizon and re-measure calibration error.	Week 7–8
5	– Complete the exogenous and causal information layer: all-variable, validation-selected and causally-filtered ablation under an identical parameter budget. – Report results separately on the direction, magnitude and risk axes. – Screen candidates for cross-pair and cross-year stability.	Week 9–10
6	– Evaluate the end-to-end economic result after measured transaction costs and slippage. – Accumulate the forecast journal and compute rolling calibration on issued forecasts.	Week 11
7	– Freeze the final configuration in writing, including variables, selection method, models, metrics and decision rules. – Prepare the sealed-set evaluation record.	Week 12
8	– Open the sealed evaluation set exactly once and score the frozen configuration. – Report results without modification and without retuning.	Week 13
9	– Complete the system: computation layer, versioned forecast artefacts, application programming interface, and user interface with full provenance for every displayed quantity. – Verify continuous daily operation.	Week 14
10	– Analyse model behaviour by volatility regime, by pair and by year. – Examine representative failure cases and document limitations.	Week 15
11	– Complete the thesis report. – Revise content based on advisor feedback. – Prepare for the final defence.	Week 16

Work Assignment:

Assignments	Member A	Member B
Review literature and define research scope	50%	50%
Construct the data pipeline and realized measures	50%	50%
Fix the evaluation protocol and sealed-set policy	50%	50%
Develop and benchmark the volatility forecasting layer	50%	50%
Develop the pattern-discovery and symbolic-representation branch	50%	50%

Implement multiple-testing control and stability screening	50%	50%
Conduct minimum detectable effect size and power analysis	50%	50%
Develop the exogenous, textual and causal information layers	50%	50%
Implement conformal prediction and calibration diagnostics	50%	50%
Develop the risk layer: VaR/ES backtesting and position sizing	50%	50%
Conduct the end-to-end economic evaluation	50%	50%
Execute the sealed-set validation	50%	50%
Implement the application programming interface and user interface	50%	50%
Analyse failure cases and document limitations	50%	50%
Write and revise the thesis report	50%	50%
Prepare for the final defence	50%	50%

Work is split evenly across all activities. Both members are jointly responsible for every stage rather than owning separate stages independently, consistent with the collaborative nature of the research.

Advisor's approval <i>(Signature and full name)</i> NGUYEN DINH THUAN	Ho Chi Minh City, ... / ... / Students <i>(Signatures and full names)</i> NGUYEN QUOC HUY HUYNH TRAN QUOC HUY
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